

Angles, polygons, congruence and vectors - Set 1

TMUA practice questions by topic. Specification M5.1-M5.6, M5.10, M5.14, M5.18, M5.19. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

1 The interior angle of a regular polygon is 156° . How many sides does it have?

- A** 24
- B** 20
- C** 10
- D** 12
- E** 18
- F** 15

2 A polygon has interior angles that sum to 1980° . How many sides does it have?

- A** 10
- B** 15
- C** 14
- D** 11
- E** 13
- F** 12

3 $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$. Find $2\mathbf{a} - \mathbf{b}$.

- A** $\begin{pmatrix} 5 \\ -5 \end{pmatrix}$
- B** $\begin{pmatrix} -7 \\ 6 \end{pmatrix}$
- C** $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$
- D** $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$
- E** $\begin{pmatrix} 7 \\ -2 \end{pmatrix}$
- F** $\begin{pmatrix} 7 \\ -6 \end{pmatrix}$

- 4 Two similar triangles have corresponding sides of length 6 cm and 9 cm. The smaller triangle has area 20 cm^2 . Find the area of the larger triangle.
- A 30 cm^2
 - B 67.5 cm^2
 - C 40 cm^2
 - D 45 cm^2
 - E 27 cm^2
 - F 60 cm^2
- 5 The triangle with vertices (1, 1), (3, 1), (1, 4) is enlarged with scale factor -2 , centre the origin. Which of the following is a vertex of the image?
- A $(-1, -4)$
 - B $(6, 2)$
 - C $(-2, -8)$
 - D $(-6, -8)$
 - E $(-3, -1)$
 - F $(2, 8)$
- 6 Which of the following sets of information is NOT sufficient to prove that two triangles are congruent?
- A two sides and a non-included angle equal (SSA)
 - B two sides and the included angle equal (SAS)
 - C two angles and a corresponding side equal (ASA)
 - D right angle, hypotenuse and one other side equal (RHS)
 - E three sides equal (SSS)
- 7 In the diagram (not shown), AB is parallel to CD . A transversal makes an angle of $(3x + 10)^\circ$ with AB and the co-interior angle with CD is $(2x - 5)^\circ$. Find x .
- A 25
 - B 55
 - C 15
 - D 30
 - E 35
 - F 45

- 8** Which of the following statements is/are true?
- I** Every rhombus is a parallelogram.
 - II** Every rectangle is a rhombus.
 - III** The diagonals of a kite are perpendicular.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III
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- 9** A shape is reflected in the line $y = x$ and then rotated 90° anticlockwise about the origin. The point $(3, 1)$ ends up at
- A** $(1, 3)$
- B** $(-3, 1)$
- C** $(-1, 3)$
- D** $(1, -3)$
- E** $(3, -1)$
- F** $(-1, -3)$
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- 10** In triangle ABC , angle A is twice angle B , and angle C is 20° more than angle A . Find angle B .
- A** 64°
- B** 30°
- C** 32°
- D** 36°
- E** 40°
- F** 28°

- 11** In triangle ABC , D lies on AB and E lies on AC with DE parallel to BC . $AD = 4$, $DB = 6$ and $DE = 5$. Find BC .
- A** 12.5
B 8
C 15
D 10
E 7.5
F 12
- 12** $ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
B P is necessary but not sufficient for Q .
C P is sufficient but not necessary for Q .
D P is neither necessary nor sufficient for Q .